15. Compare lanthanides and actinides.

16. Explain why Cr^{2+} is strongly reducing while Mn^{3+} is strongly oxidizing.

17. Compare the ionization enthalpies of first series of the transition elements.

18. Actinoid contraction is greater from element to element than the lanthanoid contraction, why?

19. Out of $Lu(OH)_3$ and $La(OH)_3$ which is more basic and why?

20. Why europium (II) is more stable than Cerium (II)?

21. Why do zirconium and Hafnium exhibit similar properties?

22. Which is stronger reducing agent Cr^{2+} or Fe^{2+}?

23. The $E^0_{M^{2+}/M}$ value for copper is positive. Suggest a possible reason for this.

24. Describe the variable oxidation state of 3d series elements.

25. Which metal in the 3d series exhibits +1 oxidation state most frequently and why?

26. Why first ionization enthalpy of chromium is lower than that of zinc?

27. Transition metals show high melting points why?

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